AI and Data Science Job Market Analysis: Factors Affecting Salaries

DSA210— Final Report | Büşra Ataseven | Spring 2025–2026

1. Motivation

The AI and data science job market has expanded rapidly in recent years, creating a wide variety of roles with significantly different compensation levels. Yet it remains unclear what actually drives these differences. Is it the technical skills a candidate holds? Their seniority? The size of the company? Or the economic conditions of the country they work in?

This project was motivated by a genuine interest in answering this question with data rather than intuition. With over 10,000 job postings available and the opportunity to enrich the dataset with World Bank macroeconomic indicators, this project applies the full data science pipeline — from exploration to machine learning — to identify what truly shapes salary in the AI job market.

Central research question:
Are salaries primarily shaped by individual qualifications (experience, skills), or by external structural and economic factors (company size, country GDP)?

2. Data Source

Primary Dataset

- File: AI_Job_Market_Trends_2026.csv
- Source: Kaggle — *"AI and Data Science Job Market 2025–2026"*
- Size: 10,345 job postings, 19 variables
- No missing values — suitable for direct analysis without heavy preprocessing

Key Variables

Type	Variables
Categorical	job_title, company_size, company_industry, country, remote_type, experience_level, education_level, hiring_urgency
Numerical	salary, years_experience, job_openings
Binary (skills)	skills_python, skills_sql, skills_ml, skills_deep_learning, skills_cloud

Summary statistics show salary ranging from approximately $45,000 to $204,000, with a mean around $113,438 and substantial variability across positions.

Data Enrichment: World Bank GDP Per Capita

To extend the analysis beyond job-level variables, GDP per capita data (2023) was obtained from the World Bank (API_NY.GDP.PCAP.CD) and merged with the primary dataset using the country column.

Enrichment Details

- 2,944 entries had no exact GDP match
- Missing GDP values were filled using the median GDP
- The enrichment enabled investigation into whether country-level economic conditions contribute to salary prediction

3. Data Analysis

The analysis followed a structured three-stage pipeline:

1. Exploratory Data Analysis (EDA)
2. Statistical Hypothesis Testing
3. Machine Learning Modeling

3.1 Exploratory Data Analysis (EDA)

Salary Distribution

The salary distribution is slightly right-skewed, with most salaries concentrated in a moderate range and a smaller group of high earners extending the upper tail. This is consistent with real-world compensation patterns in the technology sector where senior and specialized positions can command significantly higher salaries.

Salary vs. Experience Level

A boxplot of salary by experience level reveals a strong and consistent gradient:

- Entry-level positions earn the lowest salaries
- Mid-level roles earn substantially more
- Senior positions receive the highest compensation

The separation between groups is visually pronounced, suggesting experience is likely a dominant predictor even before formal statistical testing.

Salary vs. Work Type

Remote, hybrid, and onsite roles exhibit highly overlapping salary distributions. Their median salaries and average compensation levels are very similar, indicating that work type alone may not independently influence salary.

Job Titles, Company Size, and Country Distribution

The dataset contains diverse AI-related job titles including:

- AI Engineer
- Machine Learning Engineer
- Data Scientist

- Business Analyst

Company sizes are distributed across Startup, MNC, and medium-scale organizations. Country frequencies are uneven, with nations such as Germany, Australia, and Canada appearing more frequently than others.

GDP Per Capita Distribution

GDP per capita values are highly right-skewed, with a small number of high-GDP outlier countries.

Correlation analysis showed:

- GDP per capita and salary have an extremely weak relationship ($r \approx 0.018$)
- Years of experience has a moderate positive relationship with salary
- ML and deep learning skills exhibit positive correlations with compensation

3.2 Hypothesis Testing

Test 1 — ANOVA: Does Experience Level Affect Salary?

Value	Result
Method	One-Way ANOVA
H_0	Mean salary is equal across all experience levels
F-statistic	3615.56
p-value	< 0.001
Decision	Reject H_0

The ANOVA result is highly statistically significant. Experience level has a major effect on salary and appears to be one of the strongest determinants of compensation.

Test 2 — T-test: Does Work Type Affect Salary?

Value	Result
Method	Independent Samples T-test
H_0	Remote and onsite salaries are equal
t-statistic	-0.46
p-value	0.645
Decision	Fail to Reject H_0

The test indicates no statistically significant salary difference between remote and onsite positions. This suggests that work arrangement alone does not meaningfully affect compensation in this dataset.

3.3 Machine Learning

Two regression models were trained to predict salary:

- Linear Regression
- Random Forest Regressor

Features Used

- experience_level
- years_experience
- company_size
- remote_type
- education_level
- skills_python
- skills_sql
- skills_ml
- skills_deep_learning
- skills_cloud

Train/Test Split

- 80% training data
- 20% testing data

Model Results (Without GDP)

Model	RMSE	R^2
Linear Regression	20,900.96	0.5613
Random Forest	21,024.15	0.5561

Both models achieved highly similar performance. The comparable R^2 scores suggest that salary relationships in the dataset are largely linear rather than strongly non-linear.

Model Results (With GDP Per Capita)

Model	RMSE	R^2
Linear Regression + GDP	20,904.65	0.5611
Random Forest + GDP	20,710.22	0.5692

Adding GDP per capita resulted in only a very small improvement in Random Forest performance (~0.01 increase in R^2), confirming the near-zero relationship observed during EDA.

This suggests that macroeconomic conditions contribute minimally once job-level variables are already included.

Linear Regression Coefficients (Top Features)

Feature	Coefficient
experience_level	+24,575
skills_deep_learning	+15,490
skills_ml	+14,749
skills_cloud	+9,900
skills_python	+556
education_level	+312
remote_type	+137
years_experience	-75
skills_sql	-201
company_size	-1,866

Why is skills_sql negative?

This does not imply that SQL reduces salary.

In multiple linear regression, correlated predictors can create unstable or counterintuitive coefficient signs due to multicollinearity. SQL is also disproportionately common in entry-level analyst roles, which typically have lower salaries than advanced AI positions.

Therefore, the negative coefficient reflects dataset structure and overlap between predictors rather than a true causal negative effect.

Random Forest Feature Importance

The Random Forest model identified the following as the strongest predictors:

1. Experience level

2. Years of experience
3. Company size

Technical skills ranked lower overall, suggesting that career progression and organizational structure outweigh individual tools or programming languages in salary determination.

4. Findings

The EDA, hypothesis tests, and machine learning models consistently point toward the same conclusion:

Salary in the AI and data science job market is primarily determined by career progression and structural job factors — not by macroeconomic conditions alone.

Key Findings

- Experience dominates salary prediction.
  Both ANOVA and Random Forest feature importance confirm that seniority is the strongest determinant of compensation.
- Remote work does not provide a salary premium.
  The t-test found no statistically significant difference between remote and onsite salaries.
- Advanced AI skills add more value than general technical skills.
  Deep learning and machine learning contribute more strongly to salary prediction than general tools such as SQL or Python.
- GDP per capita contributes minimal predictive value.
  Country-level economic indicators do not substantially improve salary prediction once job-level variables are included.
- The dataset exhibits a largely linear structure.
  Linear Regression performed similarly to Random Forest, suggesting that salary patterns are relatively stable and interpretable.

5. Limitations and Future Work

Limitations

- The analysis relies on a single Kaggle dataset and may not generalize globally
- Important real-world factors are absent:
  - negotiation outcomes
  - company-specific compensation policies
  - visa constraints
  - cost of living differences
- Skill variables exhibit multicollinearity
- Approximately 44% of salary variance remains unexplained

- The dataset reflects job postings rather than accepted salaries

Future Work

Future extensions could include:

- Adding cost of living indices
- Applying advanced models such as XGBoost
- Investigating interaction effects between variables
- Analyzing combinations of technical skills
- Using longitudinal datasets to study salary changes over time
- Collecting real accepted salary data instead of job postings

6. Tools and Technologies

Category	Tools
Programming Language	Python 3
Data Manipulation	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Statistical Testing	SciPy
Machine Learning	Scikit-learn
Environment	Jupyter Notebook
Version Control	Git, GitHub

7. How to Reproduce

Clone the Repository

```
git clone https://github.com/Busrataseven/DSA210-Project.git
cd DSA210-Project
```

Install Dependencies

```
pip install -r requirements.txt
```

Run the Notebooks in Order

1. EDA & Hypothesis Testing

jupyter notebook "AI_&_Data_Science_Job_Market_Analysis_Factors_Affecting_Salaries.ipynb"

2. Machine Learning

jupyter notebook ML_Analysis.ipynb

AI Usage Disclosure

AI tools (ChatGPT and Claude) were used during this project to support:

- debugging Python code
- resolving preprocessing issues
- improving report structure
- refining written explanations
- reviewing interpretation clarity

All analyses, implementations, findings, and conclusions were manually reviewed and verified by the author.